Two views of dispersive wave packets

A random envelope and a pair of overlapping packets in three spatial dimensions

Abstract. We compare two artificial initial fields evolved by the same homogeneous cubic dispersive operator: a seeded random packet and a pair of phase-offset Gaussian packets. Three-dimensional level surfaces and planar slices show how their shapes change between $t = 0$ and 8. Both sampled envelope peaks fall, although their relative reductions differ. A supplementary time series separates this visible spreading from the fractionally weighted diagnostic in the proposed decay inequality. The results are finite-grid observations and support an exploratory comparison; they do not identify an asymptotic exponent.

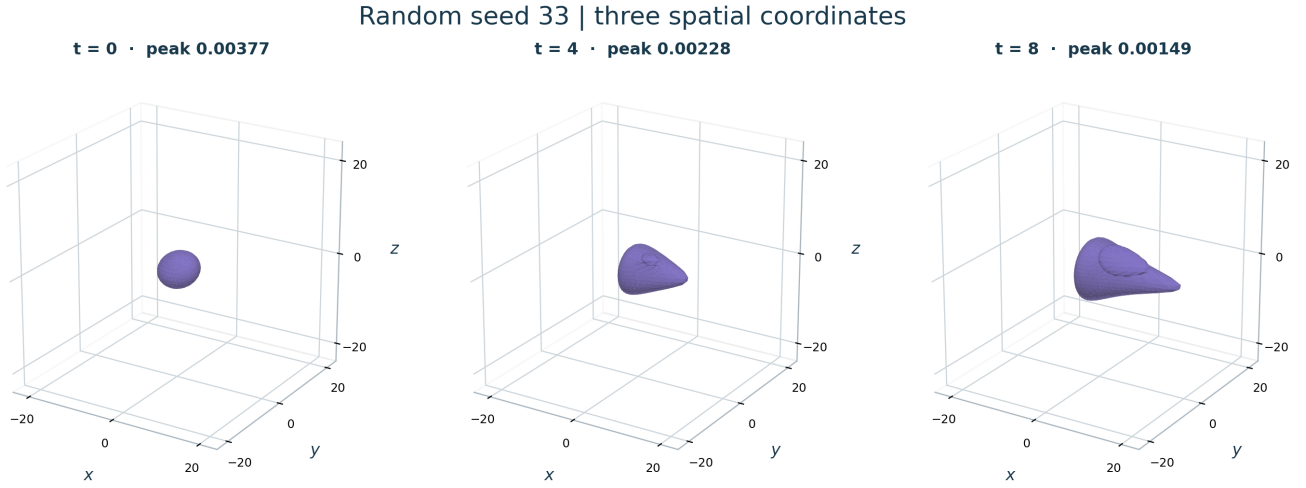


Figure 5. Random seed 33: the envelope as a volume. These are surfaces of $|u| = 0.3 \max |u|$, using the full sampled maximum separately at each time. All axes are spatial coordinates, and every panel uses the same $[-24, 24]^3$ viewing cube. The magnitude is represented by the surface threshold, not by a vertical elevation axis. The numerical maxima appear above the panels.

The initially compact central region becomes elongated and asymmetric. The geometry changes while the sampled maximum drops from approximately 0.00377 to 0.00149. Because the threshold also falls, the growing visible region cannot be interpreted as an increase of amplitude or as a quantitative measure of volume.

Main observation. The two selected fields illustrate different routes to a lower peak: deformation of a random envelope and overlap of two phase-offset packets. Their figures describe shape; the diagnostic comparison on page 5 measures amplitude and its relation to the decay question.

CASE A / LOOKING INSIDE A RANDOM ENVELOPE

From a compact magnitude profile to an asymmetric packet

The first input combines twelve complex plane waves under a Gaussian window:

$$f_{33}(X) = \frac{e^{-|X|^2/18}}{Z_{33}\sqrt{12}} \sum_{j=1}^{12} e^{i(q_j \cdot X + \phi_j)}, \qquad X = (x, y, z).$$

The wavevectors are sampled from independent centered normal components with standard deviation 0.25; phases are uniform on $[0, 2\pi)$. NumPy seed 33 fixes the realization. The positive constant Z_{33} normalizes the sampled L^1 norm to one. This is one reproducible realization, not a statistical ensemble average.

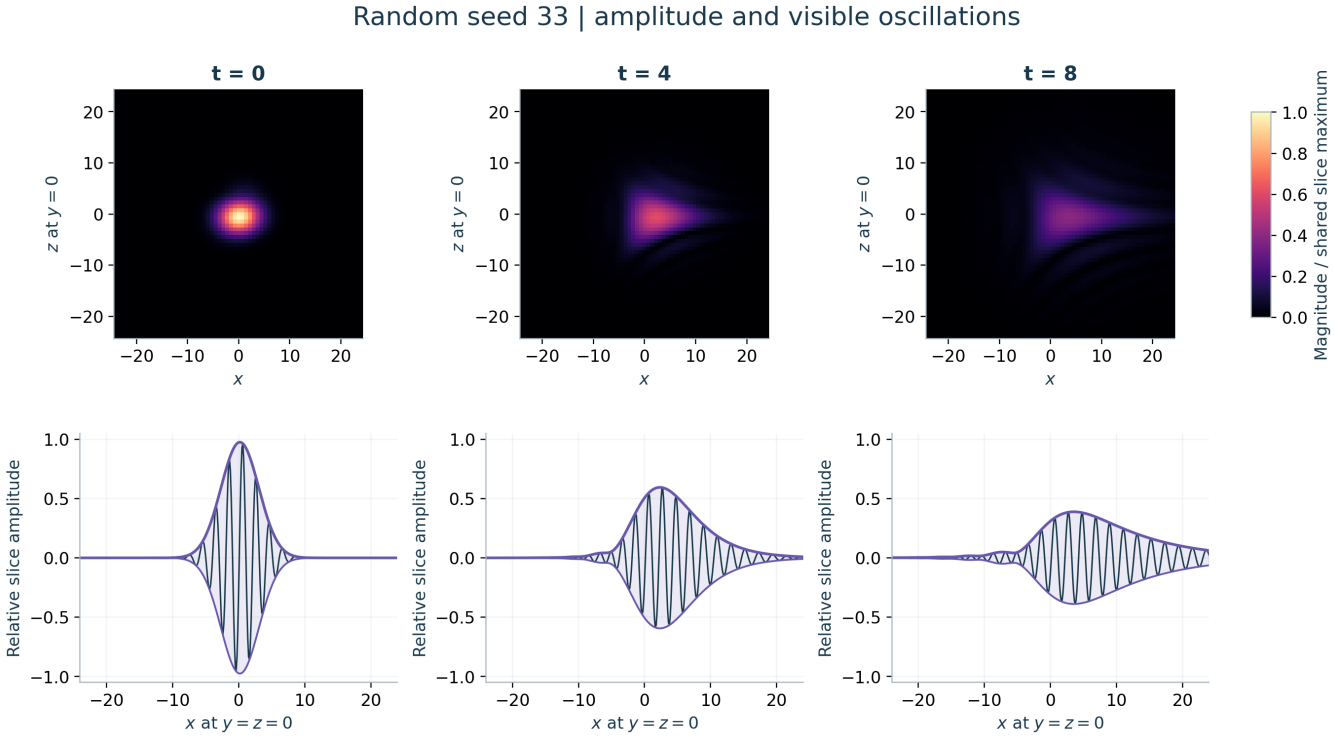


Figure 6. Random seed 33: amplitude and oscillations. Top panels show $|u(x, 0, z, t)|$. Their shared color scale divides by the largest displayed slice magnitude across the three times. Bottom panels show the centerline envelope $\pm|u(x, 0, 0, t)|$ and the illustrative carrier reconstruction $\mathrm{Re}\{u(x, 0, 0, t)e^{3ix}\}$, divided by the same constant.

The fixed color scale makes the reduction of the bright central region visible. The later slices also reveal weaker oscillatory structure away from the center. Those lower-amplitude features are largely absent from the 30% level surfaces in Figure 5: a slice and a level surface answer different questions.

The dark oscillations remain inside the positive and negative envelope. Their locations depend on the phase of the complex solution as well as on the added carrier. The complex centerline is Fourier-interpolated from spacing 0.75 to 0.125 for drawing; this makes the reconstruction smooth without adding simulated Fourier modes. The carrier wavenumber 3 is an illustrative choice.

Reading the magnitude. It is dimensionless envelope amplitude, not velocity. Neither these oscillations nor the 3D level surfaces represent a calibrated water elevation. A physical surface reconstruction would require a specified water-wave model and physical scales.

CASE B / TWO PACKETS WITH A RELATIVE PHASE

Overlap changes the combined shape

The second input is deliberately simpler than the random realization:

$$f_{\text{pair}}(X) = \frac{1}{Z_{\text{pair}}}\left[e^{-((x+5)^2+y^2+z^2)/18} + 0.7i\, e^{-((x-5)^2+y^2+z^2)/18}\right].$$

The packet centers begin at $x = -5$ and $x = 5$. Each has width 3; the second has relative amplitude 0.7 and a phase offset of $\pi/2$. As before, Z_{pair} is chosen using the grid quadrature so that the initial L^1 norm is one.

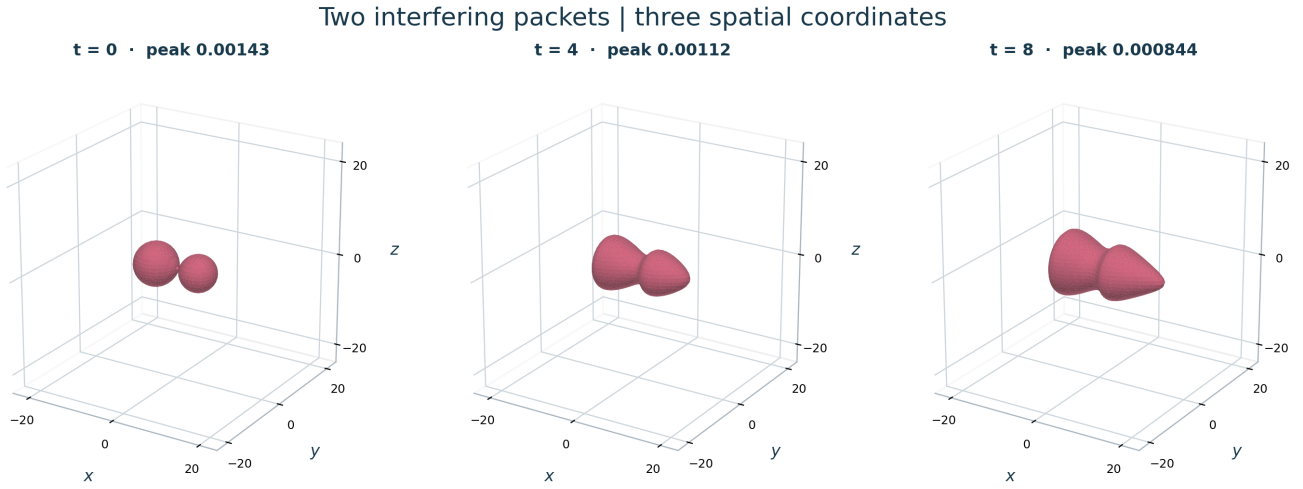


Figure 7. Two packets: evolving 3D level surfaces. The same 30%-of-current-peak rule and spatial viewing cube are used as in Figure 5. The separated initial regions develop an overlapping, asymmetric shape. The plots show thresholded magnitude geometry; they do not track individual fluid parcels.

The sampled peak decreases from 0.00143 to 0.000844, a smaller relative drop than for the random packet over the same interval. Both inputs have $L^1 = 1$, but they have different initial peaks, L^2 norms, and spectra. Equal L^1 normalization therefore does not make their initial pointwise amplitudes equal.

Why phase matters even in a linear model. Let u_1 and u_2 denote the evolved, normalized components, including the factor $0.7i$ in u_2. Linearity gives

$$u = u_1 + u_2, \qquad |u|^2 = |u_1|^2 + |u_2|^2 + 2\,\mathrm{Re}(u_1\overline{u_2}).$$

The last term is interference. Initially the Gaussian components have a quadrature phase offset, so that term vanishes. Their subsequent complex evolution can make it nonzero. The overlapping magnitude thus need not be the sum of the two separate magnitudes.

Interpretation. This is linear superposition with interference. The apparent joining of level surfaces is not evidence of nonlinear attraction, soliton interaction, or irreversible merger.

CASE B / CONNECTING THE VOLUME TO THE WAVE TRACE

The slice explains what the level surface leaves out

The planar and centerline views show the same two-packet computation from a more familiar perspective. The two initial amplitude peaks become less distinct as their evolved fields overlap, while the carrier oscillations remain visible.

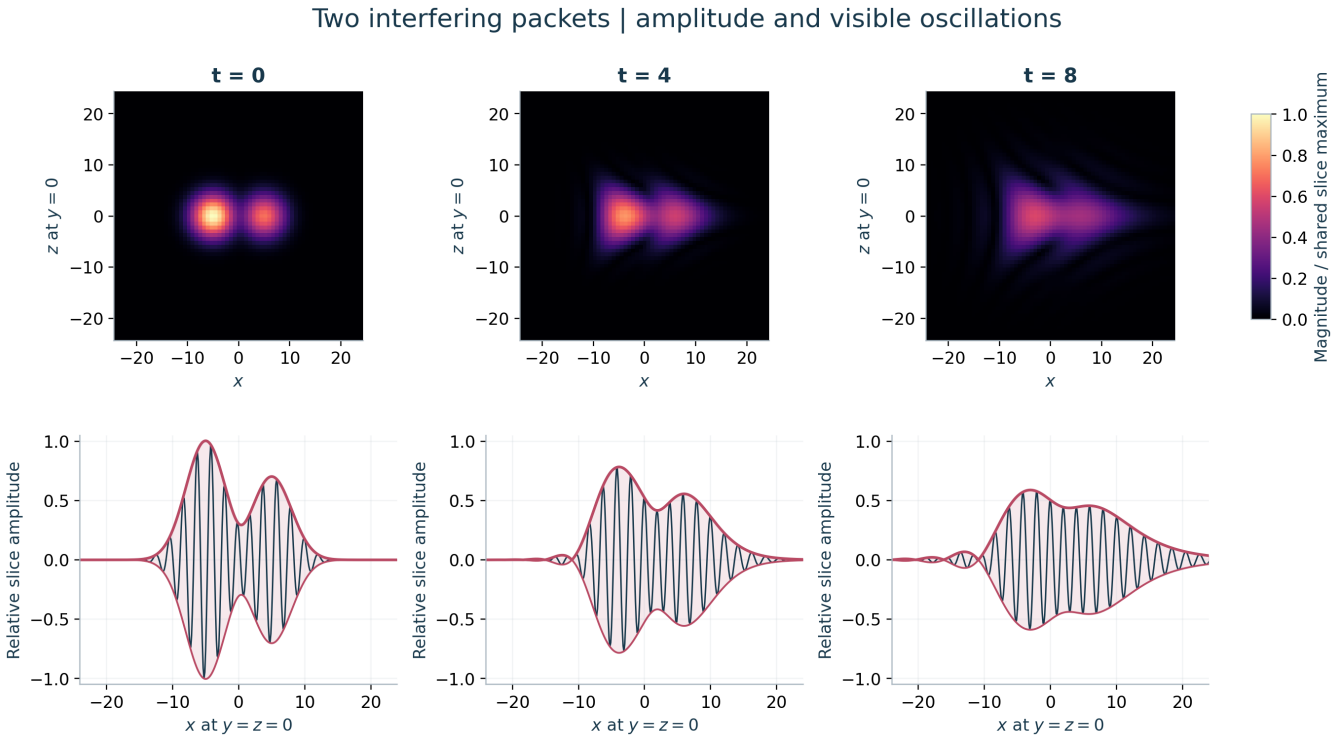


Figure 8. Two packets: fixed-scale slices and carrier reconstruction. The conventions are identical to Figure 6, but the normalization divisor is computed separately for this case. Colors can be compared across times within this figure; they cannot be used for an absolute amplitude comparison with Figure 6. The printed full-volume peaks and the table on page 5 provide that comparison.

The upper panels retain the weaker structures below the threshold used in Figure 7. The bottom row displays $\pm|u|$ as an outline rather than as an additional physical wave. The dark curve combines the simulated phase with the illustrative carrier; its crests are not numerical grid markers or tracked particles.

What the comparison reveals. The random realization begins with a more concentrated peak and loses a larger fraction of that peak. The two-packet profile spreads and overlaps while retaining a broader region of visible amplitude. These differences persist despite identical evolution rules and initial L^1 normalization. Initial phase and spectral content are therefore essential parts of the experiment definition.

What the comparison cannot decide. The figures do not determine a universal decay constant, establish monotonicity for arbitrary data, or predict which profile will have the larger compensated ratio at much later times. A centerline may also miss an off-axis maximum. The following diagnostics therefore search the entire simulated volume rather than one selected slice.

A useful pairing for the final report. Figures 5 and 7 establish the three-dimensional geometry. Figures 6 and 8 explain the amplitude and phase behind that geometry. Keeping each volume next to its companion slice avoids asking one visualization to communicate both shape and decay.

SUPPLEMENT / CONNECTING THE FIGURES TO THE DECAY QUESTION

A falling envelope peak is only one diagnostic

For the same two inputs and the same grid, we evaluate 17 times from 0 to 8. The weighted field and compensated ratio are

$$W(t) = |D_x|^{1/2} S(t) f, \qquad R(t; f) = \frac{t^{7/6} \|W(t)\|_\infty}{\|f\|_1}.$$

These are different from the unweighted envelope magnitudes shown in Figures 5–8. The ratio is displayed only for $t > 0$.

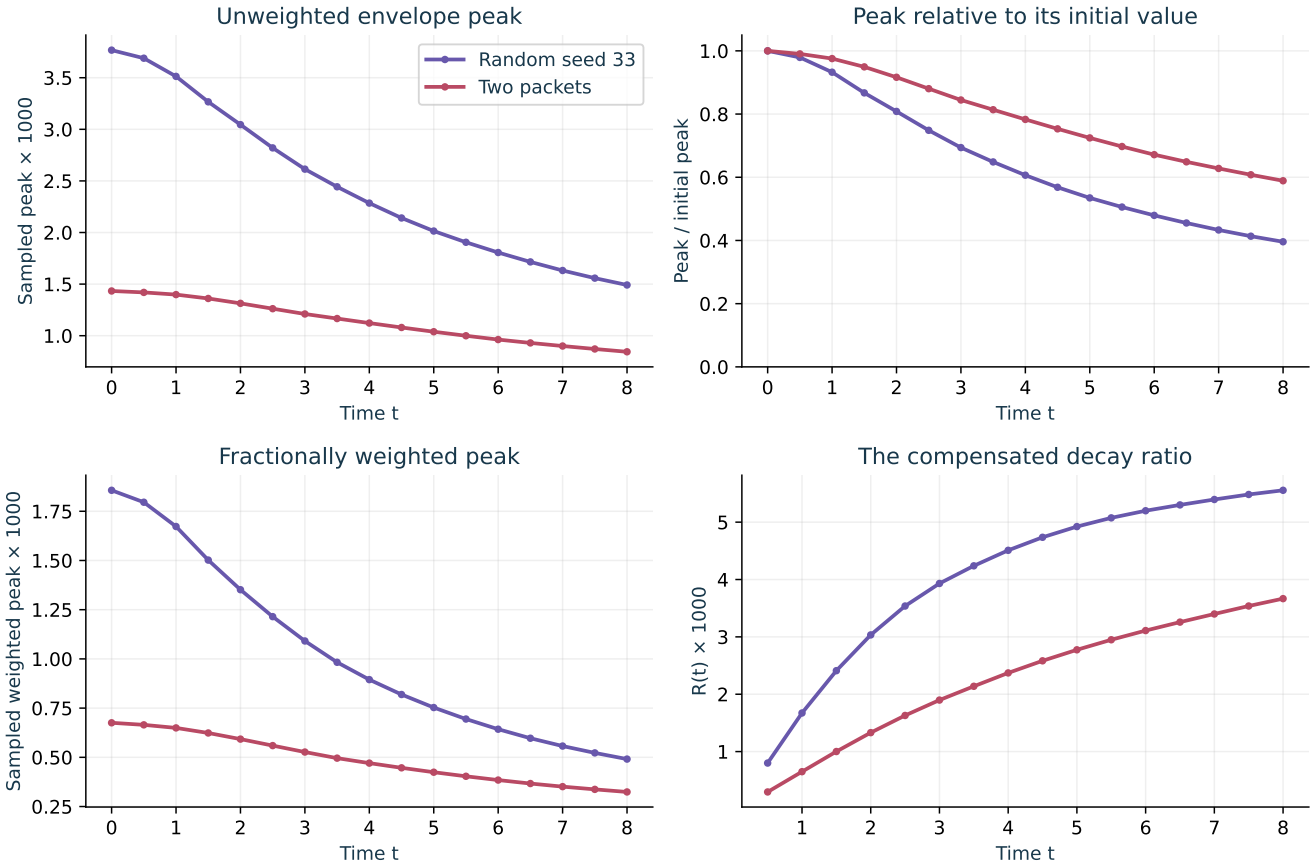


Figure S1. Same inputs, four complementary measurements. Maxima are sampled over the full periodic volume. The upper right panel removes the difference in initial peak height. The lower right panel measures the quantity associated with the manuscript's $\alpha = 1/2$, $p = \infty$ bound.

Input	Peak at 0	Peak at 8	Peak reduction	$R(8)$
Random seed 33	0.003767	0.001491	60.4%	0.005556
Two packets	0.001433	0.000844	41.1%	0.003667

The relevant distinction. A decreasing unweighted peak can coexist with an increasing compensated ratio. The proposed inequality asks whether $R(t; f)$ stays below an input-independent constant, not whether every finite curve decreases.

All curves on this page are measurements on one finite grid. Their purpose is to connect the selected visuals to the research question; they are not a new convergence study or an asymptotic fit.

METHODS, RELIABILITY AND CONTEXT

What was computed, and how to use this draft

Evolution. We work directly in the homogeneous canonical coordinates of the supplied manuscript, with

$$\widehat{u}(k,t) = e^{itw(k)}\widehat{f}(k), \qquad w(k) = k_x^3 - 3k_x(k_y^2 + k_z^2).$$

Equivalently, $u_t + u_{xxx} - 3u_{xyy} - 3u_{xzz} = 0$. The Fourier convention is $\widehat{f}(k) = \int f(X)e^{-ik\cdot X}\,dX$. This is the linear propagator used in the homogeneous weighted estimate; it is not the full nonlinear Dysthe equation.

Setting	Value
Computational domain	Periodic cube $[-72, 72)^3$
Grid	192^3, spacing 0.75
Displayed region	$[-24, 24]^3$; slices at $y = 0$
Snapshot times	$0, 4, 8$
Supplementary time series	$0, 0.5, \ldots, 8$
Arithmetic	Torch CPU, complex128
Initial normalization	Discrete $L^1 = 1$ for each input
Time evolution	Exact Fourier phase for retained modes

Checks and their limits. The new diagnostic series reproduces the three selected snapshot times. Integrated squared magnitude is conserved to numerical precision: the relative L^2 changes are below 10^{-11}. Conservation checks consistency but cannot establish an accurate whole-space maximum or detect every phase error. No separate domain enlargement or mesh refinement was performed for this selected-packet series.

At $t = 8$, the maximum sampled amplitude on the displayed cube's boundary is approximately 16.4% of the full peak for seed 33 and 6.0% for the two-packet case. Thus the 30% surfaces do not touch that sampled boundary, but weaker tails are cropped from view. This is a display check, not a test of the computational boundary. The strong $t = 256$ audit in the larger project applies to other Gaussian-based inputs and must not be transferred to these two cases.

Research context. The supplied manuscript by Mosincat, Pilod and Saut, *On Third Order Nonlinear Schrödinger Equations* (January 22, 2026), Section 6, provides the canonical symbols and weighted estimates. Its proof-phase convention is used here. Ben-Artzi, Koch and Saut's two-dimensional cubic dispersion study (2003) motivates attention to spectral geometry. For a physical free-surface model, Guyenne, Kairzhan and Sulem's Hamiltonian Dysthe study (2022) offers surface reconstruction and Euler comparisons; its fluid domain has two horizontal surface coordinates, unlike the three independent coordinates here.

Suggested scientific follow-up. First compare domain and grid changes for these exact two recipes and the weighted peaks. Then extend their observation window while tracking the global candidate maximum. Repeating seed 33 with more seeds would support an ensemble comparison; one realization cannot characterize random initial data in general.

Reproduction. Run `make_figure_review.py` for the selected figures. Run `report/selected_packet_diagnostics.py` for Figure S1, its table, and the JSON records in `results/selected_packets/`. Compile `report/selected_packets_mock.tex` to rebuild this draft.